Where 4-Bit Quantization Breaks Small Tool-Calling Agents, and Which Repairs Fix It

CMPE 295A · Advisor review · October 30, 2026

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M1 status · responses to the six points from October 16 · first cluster results · plan to M2

Progress since October 16

252 / 253

M1 skeleton episodes from one command (BFCL on CUDA and CPU, τ^2 airline, R1–R3); every step logged and classified

3 GPU types

Cluster live since Oct 26: llama.cpp built for P100, A100, H100; tool-call smoke test passes on all three

31 vs 5.9

Missing calls per 100 steps, Q4_0 vs BF16: Qwen3-0.6B on all 200 BFCL multi-turn tasks (1,600 episodes on the cluster, Oct 28)

6 / 6

Advisor points from Oct 16 answered in Workbook 2, each in bold where it lands

Done, Oct 16–29

M1 walking skeleton on Colab (local=true fallback) · Slurm association fixed, cluster measured · clean-clone check passed on the cluster · first damage runs on 200 tasks · Workbook 2 with PERT chart and the six responses · PRs for interfaces and R2/R3 merged to main

Open

Clean-clone rerun under a second member's own account · human labels for the 30-step rater sample · 3 of 5 component PRs still in review · one τ^2 episode pending · simulator as an A100 job (M2)

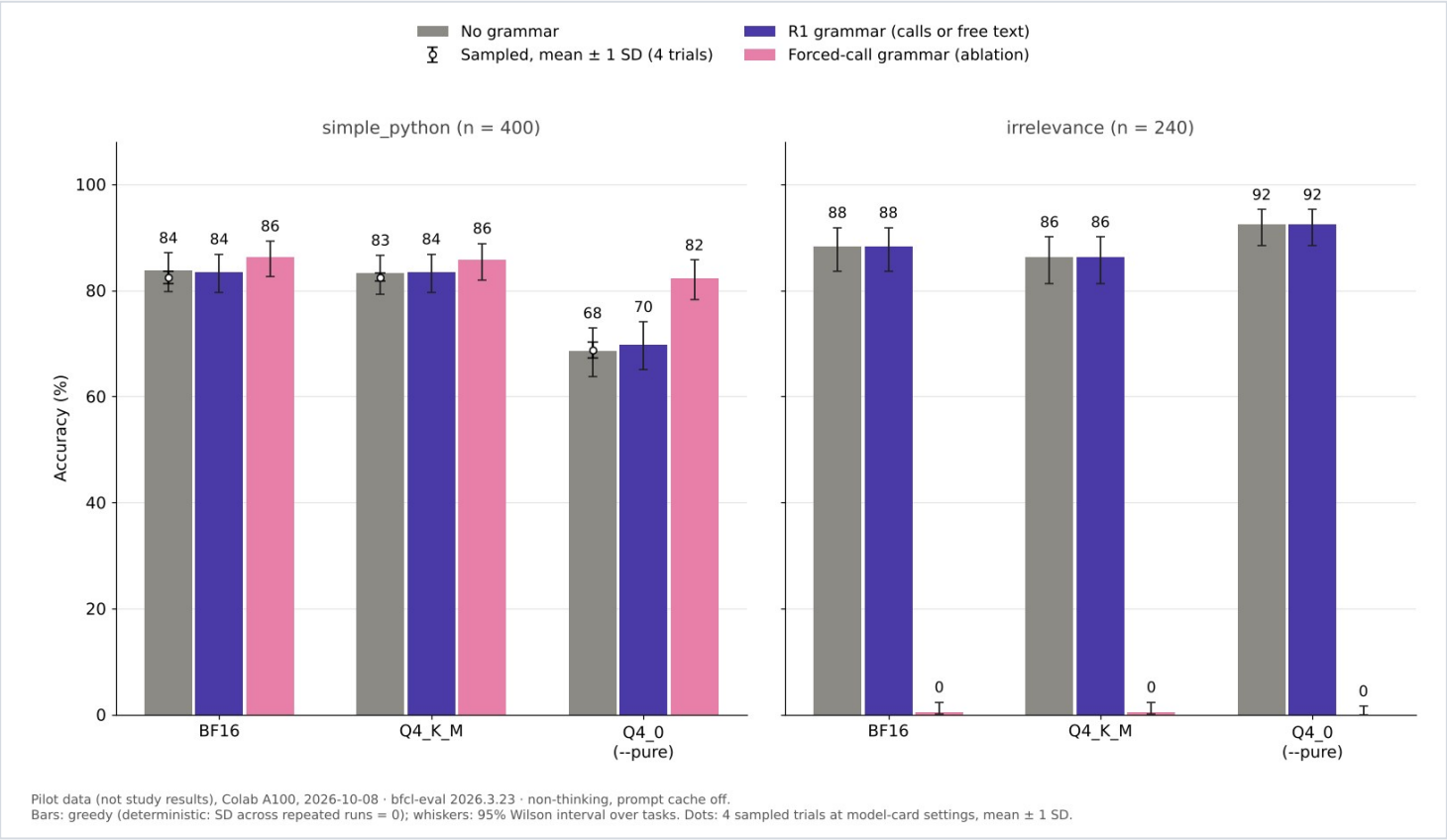
All M1 numbers are development data, not study results · Workbook 2, Chapter 8 and Table 37

Advisor points from October 16

#	Your point (Oct 16)	What we changed	Where
1	SD on every bar graph	± 1 SD over 4 sampled trials; greedy bars a 95% Wilson interval (SD = 0); rule in the analysis plan	Fig. 5a; notes under Tables 35a, 36a; Ch. 7
2	Mixed effects: in or out	In, as a registered analysis next to the bootstrap: three models (lme4 in R) and a rule for when they disagree	Ch. 3; Ch. 7 analysis plan
3	Rename “Project Architecture”	Ch. 5 is now “System and Infrastructure Architecture” with two views: software (Fig. 1) and deployment (Fig. 1a)	Ch. 5
4	Divide the plan by course	Scope by course: 295A ends with M4 and a launched QAD run (L1); training continues over the break (L2)	Table 38a; Fig. 6a
5	Architecture changes from the HPC	Eight measured cluster facts and the change each forced, e.g. the τ^2 simulator moves inside the cluster	Table 19a
6	Repair on another domain (CV)	Optional 295B layer: a vision-language agent with R1 and R2, go/no-go at M6	FR-O6 (Table 8)

Workbook 2 (submitted Oct 29): every change made for these points is in bold, as Canvas asks

Point 1 · Dispersion on every bar graph



Sampled runs

Mean ± 1 SD over the 4 trials, e.g. simple_python Q4_0 68.75 ± 1.57

Greedy runs

Deterministic (SD = 0), so a 95% Wilson interval over tasks: Q4_0 68.5% [63.8–72.9]

Reporting rule (Ch. 7)

Every effect gets its 95% paired-bootstrap interval; the report job fails if a figure has no error bars

What the figure shows: BFCL accuracy of each format without a grammar (grey), with the R1 grammar (purple) and with a forced-call grammar (pink, a test only). Bars: greedy run; whiskers: 95% Wilson interval over tasks; white dots: mean ± 1 SD over 4 sampled trials. Pilot data, Colab A100, Oct 8.

Point 2 · Mixed-effects analysis

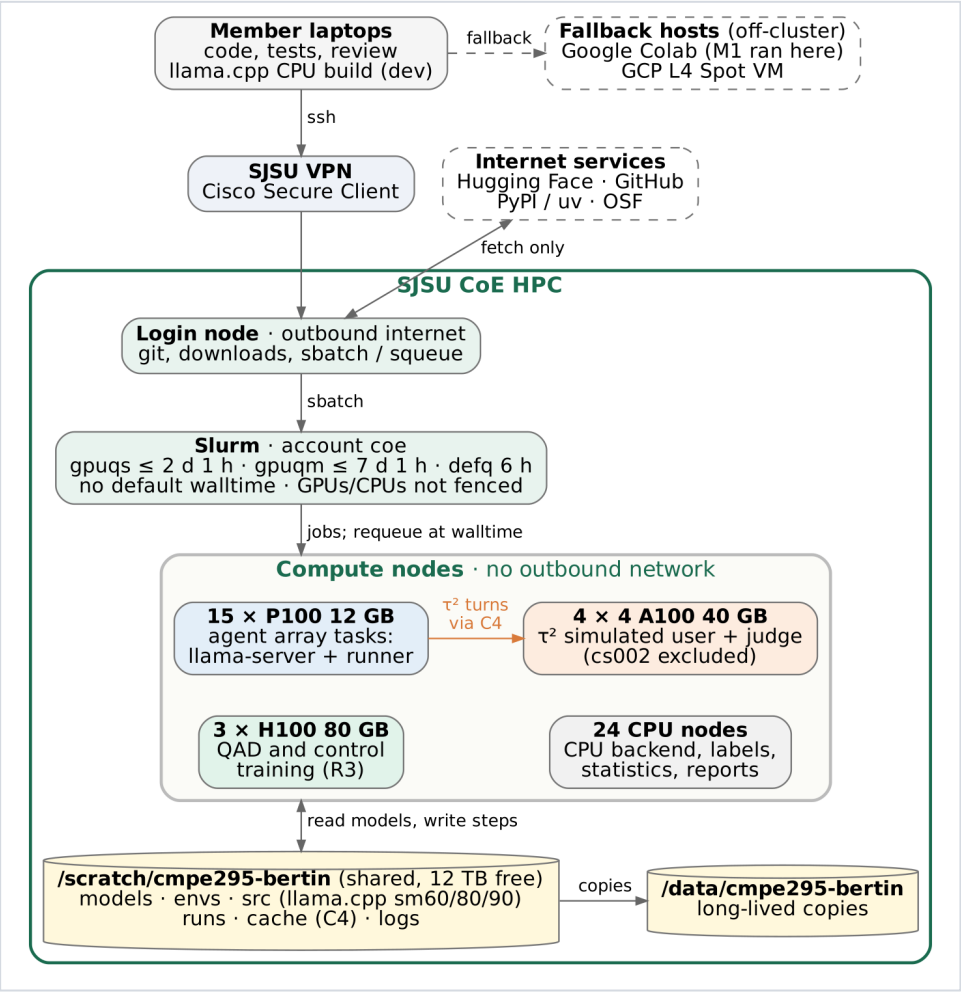
Model	Formula (lme4::glmer in R)	Answers
A · Poisson	failures ~ configuration + (1 | task), offset log(steps); negative binomial if overdispersed	Failure rate ratio per configuration
B · Binomial	failure ~ precision × step type + precision × turn depth + (1 | task/episode)	Where in an episode 4-bit damage appears
C · Binomial	success ~ configuration + (1 | task)	Episode success

Decision rule: a conclusion is robust only when the mixed model agrees with the primary bootstrap in direction and in Holm-corrected significance; otherwise we report it as not robust, with both estimates.

Why R: statsmodels has no frequentist binomial or Poisson mixed model · convergence fallbacks: bobyqa optimizer, then drop the episode intercept, then report the failure · R script and planted-effect check due at M2, frozen in the OSF registration

Workbook 2, Ch. 3 (bullet no longer “optional”) and Ch. 7 Statistical Analysis Plan

Point 3 · System and infrastructure architecture



Chapter 5 retitled

Two views: software (Fig. 1, components C1–C11 and data flow) and infrastructure (Fig. 1a, where each part runs)

15 × P100 12 GB — Agent array tasks: llama-server + runner

4 × 4 A100 40 GB — τ^2 simulated user and judge (cs002 excluded)

3 × H100 80 GB — QAD and control training (R3)

24 CPU nodes — CPU backend, labels, statistics, reports

Login node only has internet

Downloads there; jobs read only from /scratch/cmpe295-bertin

Figure 1a: how members reach the cluster (VPN → login node → Slurm), the four node types and what runs on each, and the shared storage. Dashed boxes are off-cluster.

Point 4 · Scope by course: 295A and 295B

CMPE 295A · Aug 31 – Dec 20

M1 walking skeleton (reached)

M2 BF16 references (Nov 6)

M3 damage map v1 (Dec 4)

M4 R1 and R2 evaluated (Dec 18)

L1: QAD pipeline on the H100, 0.6B run launched, first checkpoint verified

Course: Workbook 1, Workbook 2, demo and report Ch. 1–2 (Dec 4)

Winter break · Dec 21 – Jan 24

L2: QAD and control training for the remaining models, unattended on the H100 with requeue

No course deliverables

CMPE 295B · Jan 25 – May 14

M5 R3 evaluated (Jan 29)

M6 costs complete (Feb 5)

M7 workshop paper (Feb 19)

M8 release v1.0 (May 14)

Optional: vision-language layer (FR-O6)

Course: 295B workbooks, final report, Expo demo

Workbook 2, Table 38a and Figure 6a · splitting the training into L1 and L2 keeps the critical path; the buffer before M8 is 19 days

Point 5 · Architecture changes from the HPC

Measured on the cluster (Oct 26)	Change to the architecture	Where
Compute nodes have no outbound network	Every setup step split: fetch on the login node, build or run in a job, offline flags	scripts/, slurm/
Jobs cannot reach an outside VM	τ^2 simulated user moves to an A100 job inside the cluster; the GCP VM becomes the fallback (planned for M2)	C4, DEP-3
No default walltime; gpuqs 2 d 1 h, gpuqm 7 d 1 h	Every job sets -t; array tasks requeue themselves 180 s before the limit and resume; QAD checkpoints every 30 min	C2, C9
GPUs and CPUs are not fenced	Threads from SLURM_CPUS_PER_TASK; GPU, node and driver logged; exclusive nodes for cost runs	hpc_env.sh, C5
Mixed GPUs and drivers (575.57, 590.44)	One llama.cpp build for sm 60/80/90; GPU type a recorded factor; parity test per GPU type	FR-E17
Home folder full; 12 TB group scratch	Models, environments, caches and runs in /scratch/cmpe295-bertin, group-shared	hpc_env.sh
At most 2 nodes per user (QOS)	One configuration per GPU on 4-GPU A100 nodes: 4 at once, ~2 s per episode (P100 ~15 s)	multi_gpu.sbatch
Node cs002 cannot start CUDA	Excluded from A100 jobs; smoke test before each batch	smoke_s1.sbatch

Workbook 2, Table 19a · scripts on branch m1/cluster

Point 6 · A second domain: vision-language agents

What (FR-O6, RQ5)

One small open vision-language model as GGUF under the same 4-bit formats, on a visual tool-calling task set, with R1 and R2 applied unchanged and the same step classes and metrics.

Why not classic computer vision

A CNN classifier has no tool calls for our repairs to fix. A tool-calling VLM reuses the testbed, the classifier and the repairs, so the comparison stays like for like.

Guardrails: no work before M5 · a one-day spike (S11) checks that llama.cpp serves the model on a P100 and that a scored task set exists · go/no-go at M6 (Feb 5) · if dropped, BFCL and τ^2 carry RQ1–RQ4 · the QAD work (K, L1, L2) is unaffected

Workbook 2, Table 8 (FR-O6) and Schedule Risks

M1 results (development data)

Configuration (Qwen3-0.6B)	BFCL multi-turn (20 tasks)	τ^2 airline (3 tasks)
BF16, CUDA	0 of 40	0 of 6
Q4_K_M, CUDA	0 of 40	0 of 6
Q4_0, CUDA	0 of 40	0 of 6
Q4_0, CPU	0 of 40	0 of 6
R1: Q4_0 + grammar	0 of 20	0 of 3
R2: Q8_0	0 of 20	0 of 3
R3: toy QAD Q4_0	0 of 20	0 of 2 (1 pending)

6 → 12 → 29

Missing calls, BF16 → Q4_K_M → Q4_0

10 → 0

Invalid-argument steps with R1

0 → 6

Malformed calls, BF16 → Q4_0

32%

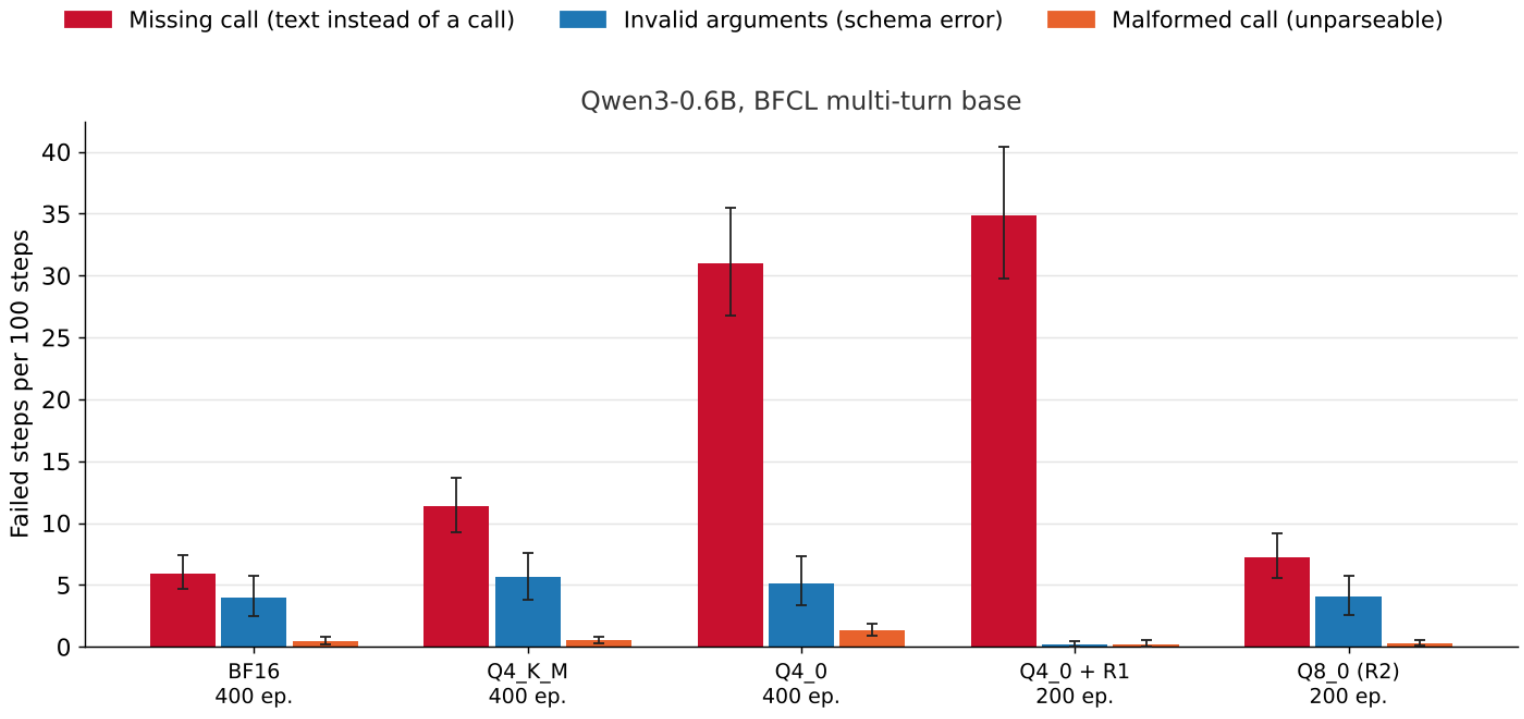
Q4_0 steps identical on CPU and CUDA

Implication for M2

Step-level outcomes become the primary outcome; backend recorded as a factor

Valid episodes out of episodes run (2 seeds) · Colab, Oct 24–26 · development data · Workbook 2, Table 36a

First damage results: Qwen3-0.6B on 200 BFCL tasks



Bars: failed steps per 100 model steps, by failure type; whiskers: 95% bootstrap CI over the 200 tasks (2,000 resamples).
BFCL multi-turn base, non-thinking, sampled decoding (T 0.7); baselines 2 seeds, repairs 1 seed; CoE HPC A100, Oct 28. Development data.

Missing calls by format

Per 100 steps: BF16 5.9 [4.7–7.4] · Q4_K_M 11.4 [9.3–13.7] · Q4_0 31.0 [26.8–35.5]

Effect of R1 (grammar)

Invalid arguments 5.1 → 0.2 per 100 steps, but missing calls stay at 34.8 [29.8–40.4] (supports H2)

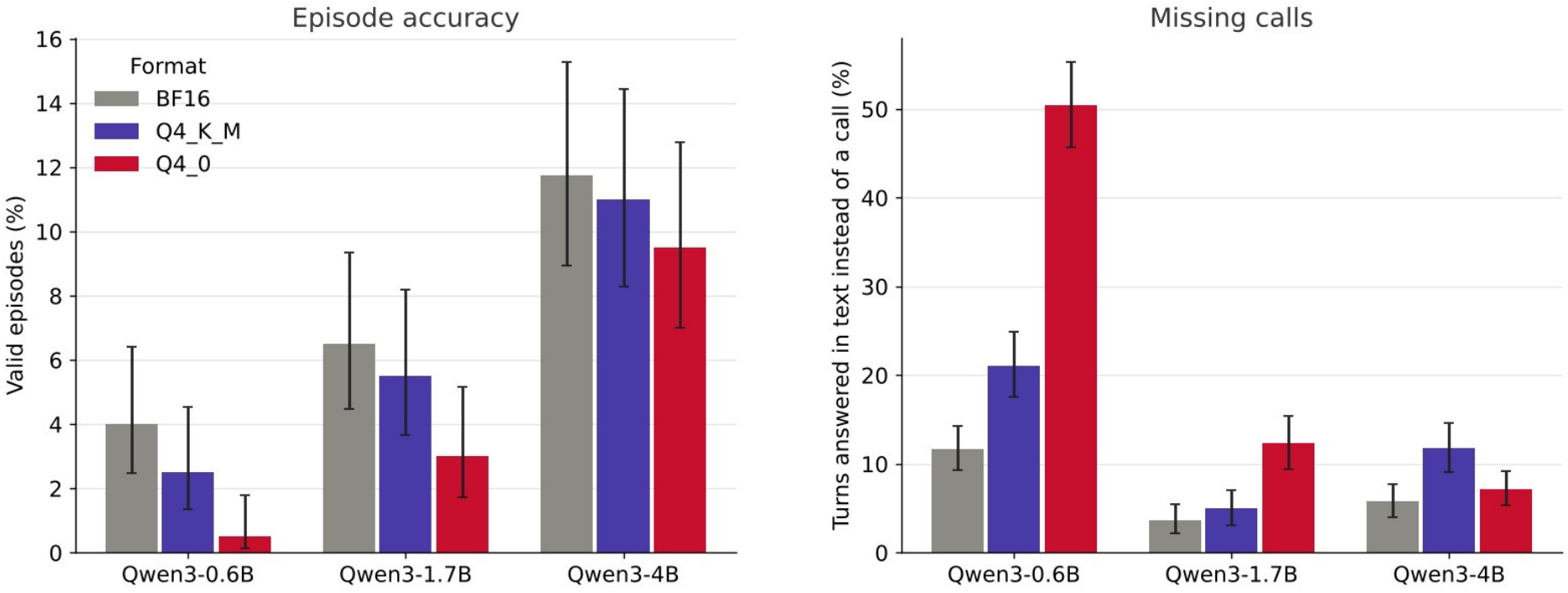
Effect of R2 (Q8_0)

Missing calls 7.2 [5.5–9.1] and invalid arguments 4.0, close to BF16

Valid episodes	BF16	Q4_K_M	Q4_0	Q4_0 + R1	Q8_0 (R2)
Qwen3-0.6B	16 / 400 (4.0%)	10 / 400 (2.5%)	2 / 400 (0.5%)	2 / 200 (1.0%)	6 / 200 (3.0%)

CoE HPC, 4 × A100 node, Oct 28 · 1,600 episodes, 10,483 steps · development data · brackets: 95% bootstrap CI over tasks

Model size: Qwen3-0.6B, 1.7B and 4B, same 200 tasks



Valid episodes, BF16 / Q4_K_M / Q4_0: 0.6B 16 / 10 / 2, 1.7B 26 / 22 / 12, 4B 47 / 44 / 38 (of 400). Turns answered in text instead of a call fall with size; Q4_0 is worst at 0.6B and 1.7B, but at 4B Q4_K_M is worst (11.8% vs 7.1% for Q4_0 and 5.8% for BF16).

Left: 95% Wilson interval; right: 95% bootstrap CI over tasks · 200 BFCL multi-turn base tasks × 2 seeds per format · CoE HPC A100, Oct 28 · development data

First results on the CoE HPC (Oct 26–27)

144 / 231 / 350

Tool-call smoke test, tokens/s on P100 / A100 / H100 (S1 passes on all three)

21 min

llama.cpp built from source for sm 60, 80 and 90 in one job; model files byte-identical to the pilot

PASS × 2

Clean-clone check: same 253-spec plan and hash, 10 Q4_0 episodes, 60 steps all classified, 0 of 10 as reference

κ = 1.00

Blind second labelling of 30 steps (6 classes) from 100 new cluster episodes, against the written codebook

How these were run

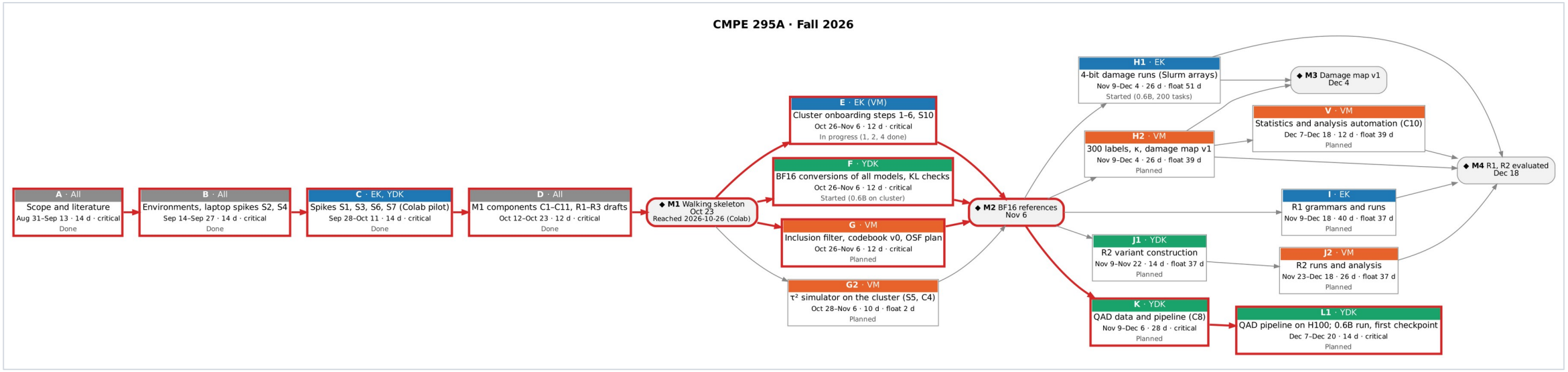
Both checks on Oct 27 ran under one team cluster account; the full rebuild was done by an independent AI agent, and the 30 labels came from an AI rater (a different model from the one that drafted the classifier). Before M2 the team repeats the check under a second member's own account and adds human labels for the same 30 steps.

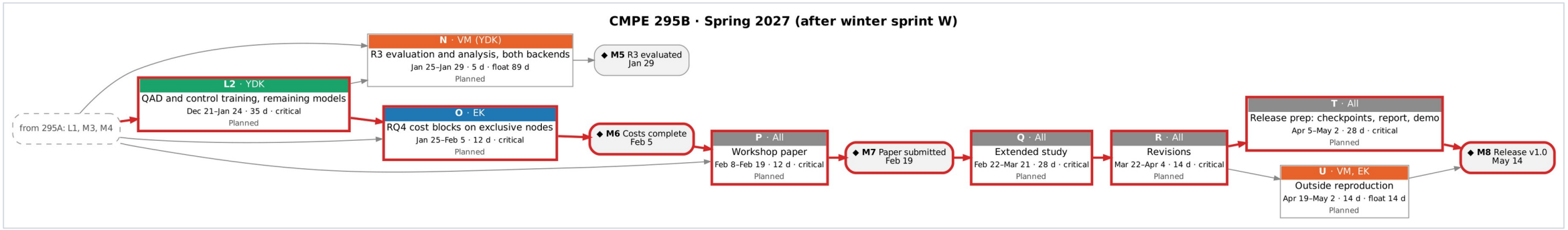
Codebook gap found

A tool that returns an error payload (e.g. "No such file") is labelled ok_call, because exec_error needs the executor's own error. Codebook v0 (M2) decides whether tool-level errors get their own class.

Slurm jobs 97498, 97528, 97611–97655 · Workbook 2, Tables 37 and 37b, Ch. 7

Schedule and critical path





EK lead VM lead YDK lead Whole team Critical path (zero float)

Each box: package, lead, dates, duration, float, status · each member leads about 104 days of work · float: damage runs 51 days, R1 and R2 37 · 19 days of buffer before M8 (May 14) · Workbook 2, Figure 6a, Table 39a

Plan to M2 (Nov 6)

Testbed & runtime

Clean-clone rerun under a second member's own account · S1 and S10 on the cluster · CUDA-vs-CPU parity test (FR-E17) · Slurm array runner with requeue

Benchmarks & analysis

Human labels for the 30-step rater sample · decide on tool-level errors · inclusion filter · codebook v0 · OSF pre-registration with the mixed-effects plan

Models & repairs

BF16 conversions of all models with KL checks · τ^2 simulator as an A100 job · mixed-effects R script and its planted-effect check

M2 on Nov 6: BF16 references for every model on the cluster, two members' own cluster runs, codebook v0, and the three remaining component branches merged to main.

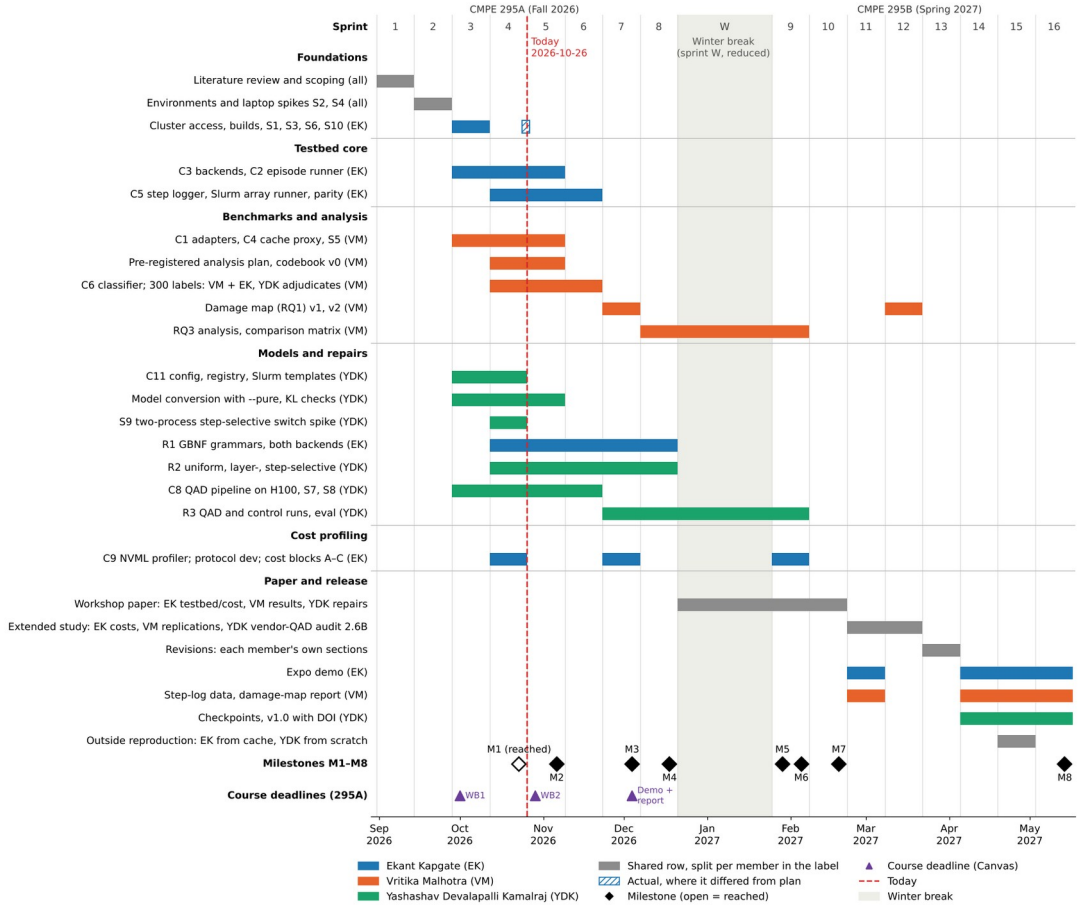
Workbook 2, Tables 37b and 39 (Sprint 5)

Decisions we need from you

1. Accept M1 as reached on Colab, with the cluster criteria and the second member's rerun at M2
2. Make step-level outcomes the primary M2 outcome (episode accuracy is 0% at 0.6B)
3. τ^2 simulator from M2: an A100 job inside the cluster (recommended), or keep the GCP VM
4. Confirm the dispersion convention and the mixed-effects analysis as on points 1 and 2
5. M4 on Dec 18 as planned, or Dec 11 so 295A closes on a finished milestone
6. Accept the vision-language extension as optional 295B scope, go/no-go at M6

Full schedule, Fall 2026 to Spring 2027

Backup



What the figure shows: every task by sprint for both semesters; bar colour = lead (blue EK, orange VM, green YDK, grey whole team); dashed line = status date Oct 26; diamonds = milestones; triangles = course deadlines. Workbook 2, Figure 6.

Where each Workbook 2 rubric item is answered

Backup

Rubric item	Where in Workbook 2
WB2-6 Evaluation Methodology	Ch. 3 baselines (Tables 11–15); Ch. 7 validity, statistical analysis and evaluation plans; Ch. 8 pilot (Table 35a) and M1 results (Table 36a)
WB2-7 System Design / Methodology	Ch. 5 System and Infrastructure Architecture (Figures 1, 1a; Tables 19, 19a, 20); Ch. 6 design; Ch. 8 reproduction steps
WB2-8 Implementation Plan and Progress	Ch. 8 (Tables 31–37b): environment, spikes, pilot, M1 results, progress log, plan to M2, cluster onboarding, decisions
WB2-9 Project Schedule	Ch. 9: Gantt (Figure 6), PERT (Figure 6a, Table 39a), milestones and scope by course (Tables 38, 38a), sprint plan (Table 39)

Workbook 2, Table 20a